



VISION SUCCESS · UNA, HIMACHAL PRADESH

FREE STUDY MATERIAL · NO. 04

JEE MAIN COMMAND CARD

The exact paper structure, the two-session rule that most aspirants waste, the elimination economics of +4/-1, and the core Physics, Chemistry and Maths formulae. From an NIT Hamirpur alumnus who sat in the seat you want.



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Free to download, free to share. Made for students of Una.

KNOW THE MACHINE

The Paper, Exactly

Format	Computer-based · 75 questions · 300 marks · 3 hrs
Per subject	Physics / Chemistry / Maths – 25 each
Question split	20 MCQ + 5 numerical value questions
Marking	+4 for correct · -1 for wrong
Sessions	Twice a year (Jan + Apr) – your BEST score counts
Next level	Top ~2.5 lakh scorers qualify for JEE Advanced
Boards rule	75% in Class 12 (or top-20 percentile) for NIT/IIIT

THE FREE REHEARSAL MOST ASPIRANTS SKIP

Only your best session counts. That makes the January attempt a full-dress rehearsal with real stakes and zero downside – sit it even at 70% preparation, and walk into April with the map already drawn.

MATHS

Core Formulae

QUADRATIC ROOTS

$$x = [-b \pm \sqrt{b^2 - 4ac}] / 2a$$

SUM / PRODUCT OF ROOTS

$$a+b = -B/A, ab = C/A$$

BINOMIAL TERM

$$T(r+1) = nCr a^{n-r} b^r$$

DERIVATIVE (PRODUCT)

$$(uv)' = u'v + uv'$$

DERIVATIVE (QUOTIENT)

$$(u/v)' = (u'v - uv')/v^2$$

INTEGRATION BY PARTS

$$\int u dv = uv - \int v du$$

STRAIGHT LINE

$$y - y_1 = m(x - x_1)$$

CIRCLE

$$(x-h)^2 + (y-k)^2 = r^2$$

PHYSICS & CHEMISTRY

The Ones Worth Reflex Speed

KINEMATICS

$$v^2 = u^2 + 2as$$

NEWTON'S SECOND LAW

$$F = ma$$

WORK-ENERGY THEOREM

$$W = dKE$$

SHM

$$T = 2\pi \sqrt{m/k}$$

GRAVITATION

$$F = G \frac{m_1 m_2}{r^2}$$

COULOMB'S LAW

$$F = k \frac{q_1 q_2}{r^2}$$

OHM'S LAW / POWER

$$V = IR, P = VI$$

LENS / MIRROR

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

MOLE CONCEPT

$$n = \frac{\text{given mass}}{\text{molar mass}}$$

IDEAL GAS

$$PV = nRT$$

GIBBS FREE ENERGY

$$dG = dH - T dS$$

PH

$$pH = -\log[H^+]$$

STRATEGY

Elimination Economics

- **At +4/-1, elimination is an investment.**

Blind guessing at 25% accuracy nets you zero. Eliminate one option and the same guess becomes mathematically profitable. Every attempt should either earn, or be skipped on purpose.

- **Previous-year questions are the real syllabus.**

NTA recycles concept patterns relentlessly. Ten years of chapter-wise PYQs cover most of what the paper can legally ask. Drill them weekly, chapter by chapter.

- **Boards and JEE are one syllabus, not two.**

Learn each concept once, then layer board answer-writing and JEE problem-solving on top. Students who separate them do double the work for worse results.

YOU HAVE THE MAP

Now Get The Guide.

This sheet is the free part. The rest – weekly tests, doubt-clearing the day a doubt appears, mock analysis, and a plan built around your calendar – is what we do in the classroom, in batches of never more than fifteen.

For JEE specifically: concept-first Physics and Maths from an NIT graduate, NCERT-anchored Chemistry, and weekly NTA-pattern mocks with error analysis.

FREE 1-ON-1 STRATEGY SESSION

A short diagnostic, an honest read of where you stand, and a written plan – target, timeline, weekly schedule. You keep the plan whether or not you join us.

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Or just message / call: +91 82192 54332

Fees at Vision Success are set on your family's ability and stay well below big-city coaching. No capable student is turned away over money – come talk to us first.

